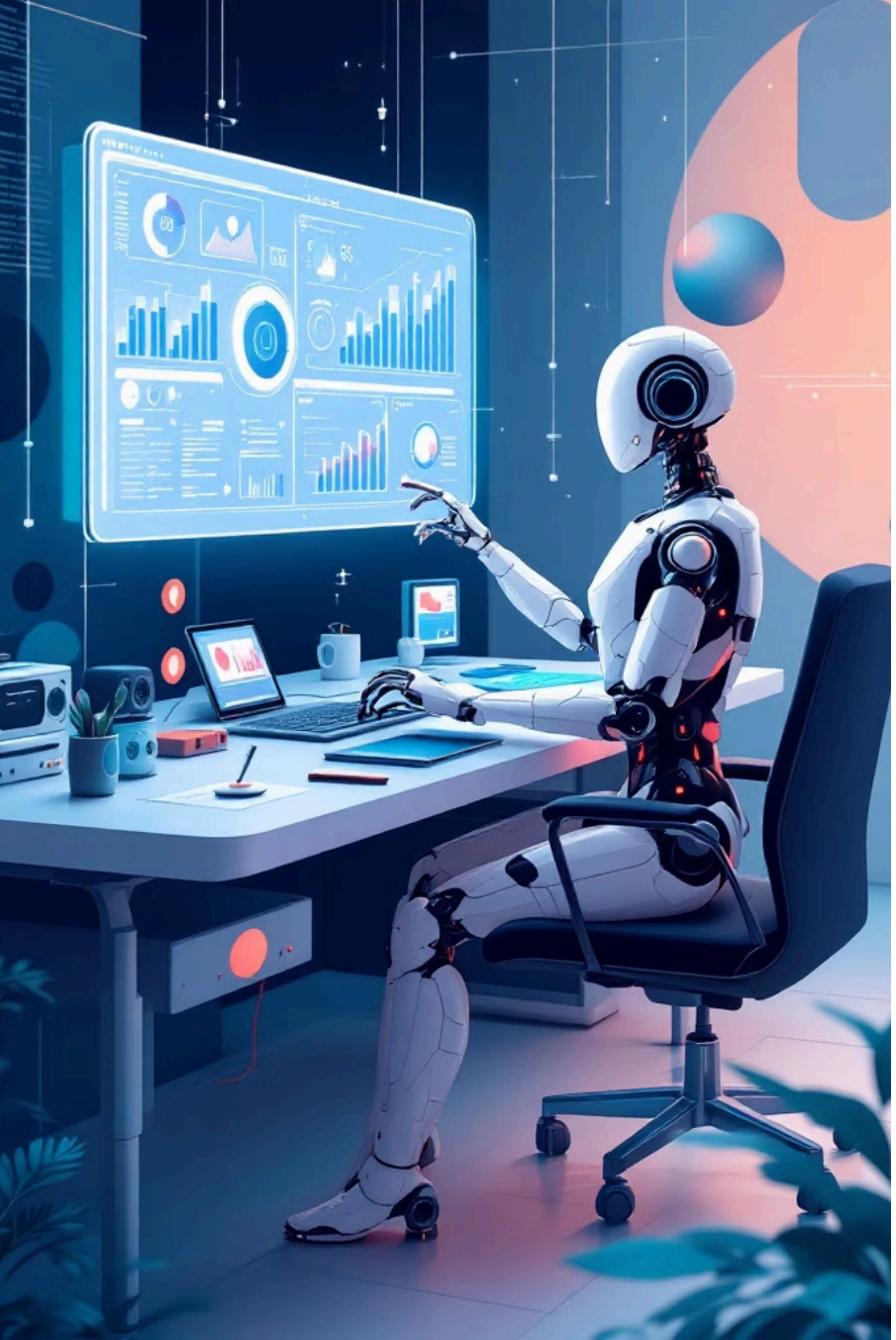


AI Agents and Tool Calling

Phase 2 – GenAI Frameworks & Building

Made with GAMMA

What are AI Agents?

AI Agents are intelligent systems that use Large Language Models to reason, plan, and take actions autonomously. Unlike simple LLM prompts, agents can interact with external tools, APIs, and systems to perform tasks automatically.

Reason & Plan

Understand problems and create step-by-step solutions

Interact with Tools

Use APIs, databases, and external systems

Take Actions

Execute tasks automatically without human intervention

Made with GAMMA

Agentic Workflow

The agentic workflow describes how AI agents process requests and solve problems through a series of steps:

01

Understand Request

Parse and comprehend the user's query or task

02

Plan Steps

Break down the problem into actionable steps

03

Choose Tools

Select appropriate external tools or APIs if needed

04

Execute Actions

Run the planned operations and gather results

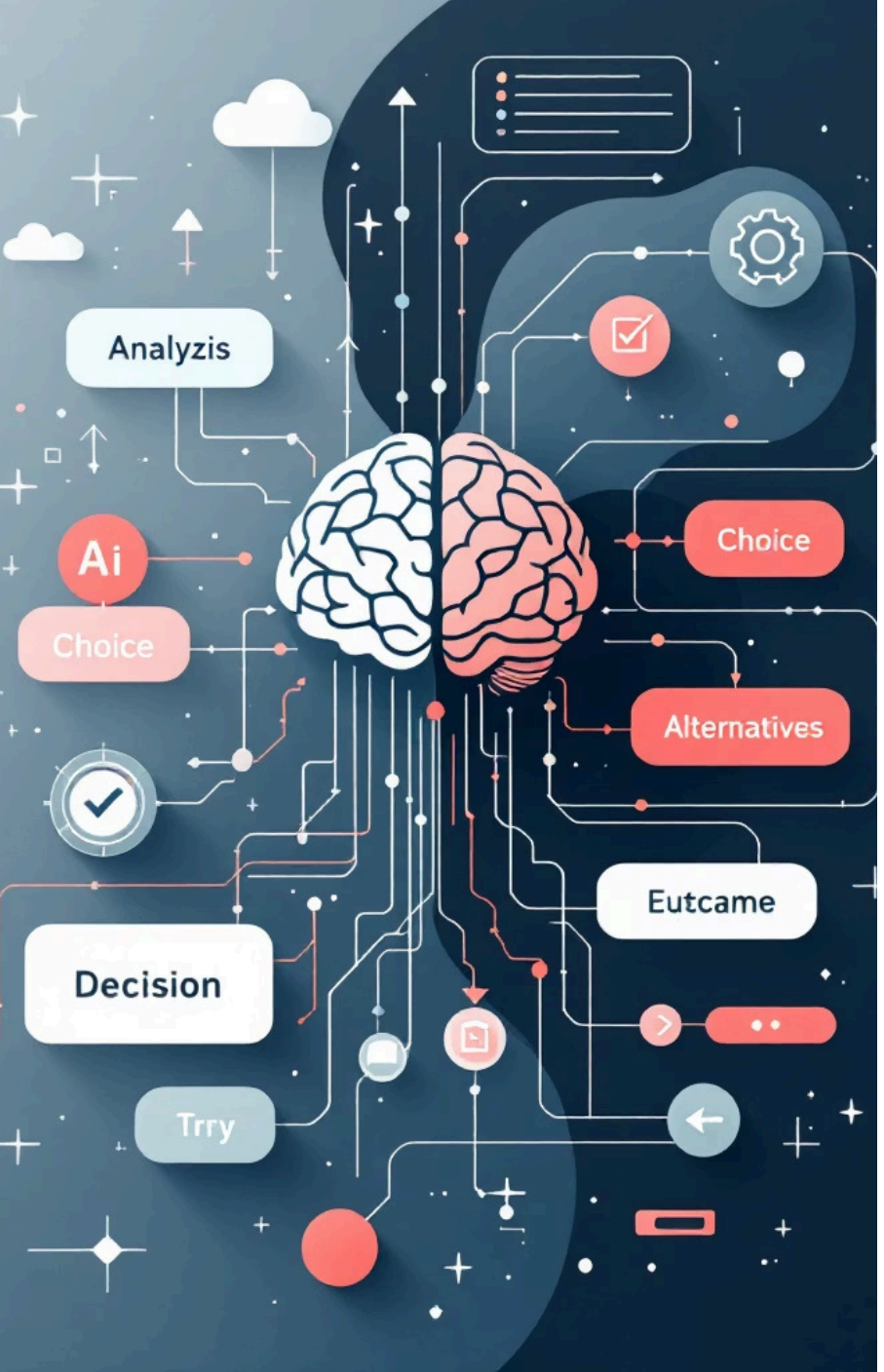
05

Return Result

Deliver the final answer or completed task

Example: User asks "Find latest AI news and summarize" → Agent searches web → Reads results → Generates summary

Made with GAMMA



Decision Making in AI Agents

AI agents use sophisticated reasoning to make decisions at each step of their workflow:

→ What action to take

Determine the next operation based on the current state and goal

→ Which tool to use

Select the most appropriate external function or API for the task

→ When to stop

Know when the task is complete or needs to be terminated

> **Common decision loop:** Think → Decide → Act → Observe → Repeat

This iterative process allows agents to solve complex, multi-step tasks autonomously.

Made with GAMMA

Tool Calling Concept

Tool calling enables LLMs to use external tools or functions to extend their capabilities beyond text generation.

Web Search APIs

Access current information from the internet

Calculators

Perform complex mathematical operations

Databases

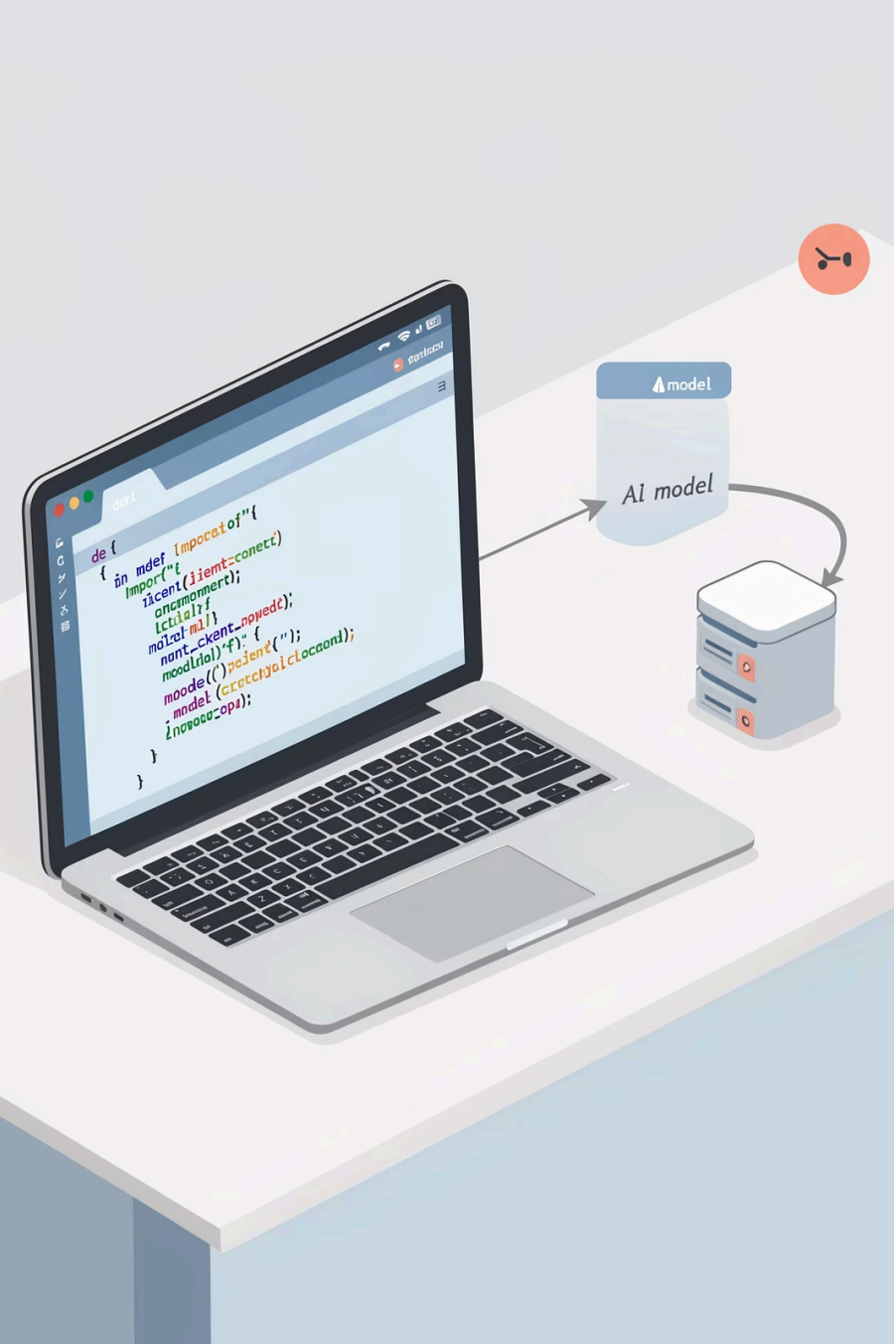
Query and retrieve stored data

Python Functions

Execute custom code and logic

Why tool calling matters: LLMs cannot access real-time data or perform computations. Tools provide updated information and additional processing power.

Made with GAMMA



Binding Python Functions to LLMs

Developers can connect Python functions as tools for AI agents. The LLM decides when to call these functions based on the user's request.

Example Function

```
def get_weather(city):
    return "Weather data for " +
city
```

The agent automatically calls this function when a user asks about weather in a specific city.

How It Works

- LLM identifies the need for weather data
- Extracts city name from user query
- Invokes get_weather function
- Processes returned data
- Generates natural language response

Made with GAMMA

Example Python Code

Creating a Tool with LangChain

```
from langchain.tools import tool

@tool
def search_topic(topic: str):
    """Search the internet for information about a topic"""
    return f"Searching information about {topic}"

@tool
def get_current_time():
    """Get the current time"""
    return "Current time data"
```

This code registers functions as tools that an AI agent can use. The agent can automatically select and call these tools when appropriate during its workflow.

@tool Decorator

Registers the function as an agent tool

Type Hints

Helps the LLM understand input parameters

Documentation

Explains what the tool does

Made with GAMMA

Research Agent Example

A research agent combines multiple tools to search, collect, and summarize information automatically.

1

User Query

"Tell me about renewable energy"

2

Web Search Tool

Finds relevant articles and sources

3

Extract Data

Collects key information from sources

4

LLM Summary

Generates concise overview

Use Cases

- Academic research
- Market analysis
- News summarization
- Competitive intelligence

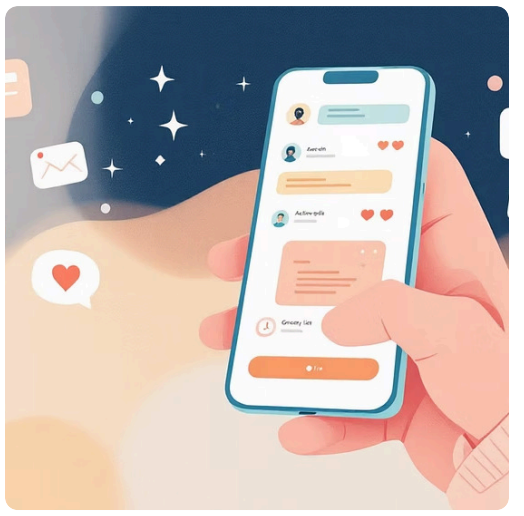
Made with GAMMA

Real World Applications

AI Agents are transforming how we work across various industries:

Customer Support

Automated assistants handle inquiries, resolve issues, and escalate complex problems



Personal Assistants

Manage schedules, send messages, and perform daily tasks through natural conversation

Coding Assistants

Generate code, debug errors, and explain programming concepts in real-time

Research Tools

Search literature, analyze papers, and synthesize findings across multiple sources

Business Analysis

Process data, generate reports, and provide strategic insights from complex datasets

AI Copilots

Context-aware assistants that enhance productivity across writing, design, and development

Made with GAMMA

Summary & Student Task

Key Takeaways

- AI agents extend LLM capabilities beyond text generation
- Agents can reason, plan, and take autonomous actions
- Tool calling enables interaction with external systems
- Agents enable powerful automated workflows

Student Task: Build Your Agent

01

Get User Input

Create a function that takes a topic from the user

02

Search Information

Implement a web search tool to find relevant information online

03

Generate Summary

Use an LLM to create a concise summary of the collected information

Hint: Use LangChain's @tool decorator to register your functions, then integrate them with an agent executor to handle the decision-making process.

Made with GAMMA